

LESSON 103 (2.0 GROUND)**Lesson Objective**

During this lesson, the student will be introduced to normal checklist procedures and how the checklist is utilized. The student will also become familiar with the flight deck layout and conduct a preflight inspection with the instructor.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the student if they got stuck at any point in their practice weight and balance problems.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Pilot's Guide to a Preflight Briefing. ([AC 91-92](#))
 - Aeronautical Decision Making. ([AC 60-22](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
- Warrior III Pilot's Information Manual.
 - Section 01 - General.
 - Section 07 - Systems.

Lesson Content**Checklist Procedures**

- Emphasize the importance of proper checklist usage and its impact on safety.
- Show location of checklists.
 - Checklists must be bought at customer service! They are also available digitally on the Florida Tech Hub.
- Distinguish between normal and emergency checklists.
- Emphasize the difference between read-and-do checklists and do-and-verify checklists.

Flight Deck Familiarization

- Take the student out to the airplane and have them sit in the flight deck.
- Be sure to point out that switches are not always in the same location depending on the airplane.
- Locate and describe basic flight deck elements and controls.

Preflight Inspection

- Emphasize the importance of a proper pre-flight inspection.
- Checklists MUST be used while pre-flying.
- Conduct a pre-flight inspection with the student.

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- Introduce AC 91-92 to the student.

Operation of Systems - Powerplant and Propeller

- PIM Section 01 (general) and Section 07 (systems).
- PHAK chapter 07 (Aircraft Systems).
- Powerplant (CL4HAND).
 - Carbureted.
 - Lycoming O-320-D3G.
 - 4 cylinders.
 - Horizontally opposed.
 - Air cooled.
 - Normally aspirated.
 - Direct drive.
 - 160HP at 2700 RPM at sea level.
 - 4 stroke engine = 2 revolutions per power stroke.
 - Exhaust system.
- Dual Ignition:
 - A magneto is an engine driven generator with a permanent magnet surrounded by coils of wire.
 - This electricity powers the spark plugs and only the spark plugs.
 - Dual ignition allows for a more complete combustion of the fuel and air mixture.
 - Completely independent of the electrical system.
- Carburetor:
 - The carburetor is responsible for mixing fuel and air in preparation to be combusted.
 - Discuss how a carburetor works through the use of the PHAK and visuals.
 - Emphasize how carburetor icing is a major weakness that causes power loss.
 - Review the temperature and humidity ranges.
 - Most likely to occur when the temperature is below 21C/70F and the relative humidity is higher than 80%.
 - Can also occur with temperatures as high as 38C/100F and a relative humidity as low as 50%.
 - Introduce how to detect and solve the icing.
- Mixture:
 - Rich:
 - High fuel to air ratio.
 - Discuss how spark plug fouling can occur due to carbon deposits.
 - An overly rich mixture results in excess fuel not being combusted.
 - Lean:
 - Low fuel to air ratio.
 - Detonation, engine roughness, overheating is possible with an excessively lean mixture.

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- Review the leaning procedures for best economy and best power (refer to the leaning service instruction on the FIT hub).
- Review the engine starting process.
- Review pre-ignition and detonation.
 - Pre-ignition: the uncontrolled firing of the fuel/air charge in advance of the spark ignition.
 - Due to hot spots (tip of the spark plug) usually in only one cylinder.
 - Detonation: explosive or instantaneous combustion of unburned charge in the cylinders
 - Due to too lean of a mixture, low fuel grade, high temperatures (usually in all cylinders).
 - Detonation can lead to pre-ignition.
- Propeller:
 - Introduce the type of propeller and limitations.
 - Fixed pitch Sensenich - 74 inches in diameter - aluminum alloy.
 - Backside painted black to reduce the sun's reflection.
 - Direct-drive: the propeller is bolted to the crankshaft.

Post-Brief

- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 104.
- Assign homework:
 - Provide the student with the PA-28-161 systems PowerPoint (Located on the training drive).
 - Review the checklists and begin to memorize all memory items and emergency checklists.
 - Prepare a TOLD card for the 104 sim using an airplane assigned by the instructor.
 - Read chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.

Completion Standards

This lesson is complete when the student is able to:

1. Understand checklist procedures to include all elements on the preflight inspection.
2. Locate and describe basic flight deck elements and controls.
3. Understand elements of the powerplant and propeller.